DhanChetna: A Context-Aware Expense Tracking System for Students

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***Abstract*—Financial management is a persistent challenge among college students owing to irregular income sources, dependency-based financial structures, and impulsive micro-spending behaviour. Existing personal finance applications are designed for salaried professionals and do not incorporate contextual modelling tailored to student lifestyles. This paper presents the design, implementation, and evaluation of *DhanChetna* (Hindi: Money Awareness), a context-aware expense tracking system for college students built as a full-stack application comprising a Django 5.2 REST API backend, server-rendered web frontend, and Tkinter desktop application. A statistically guided synthetic dataset of 412 simulated users across four living situations and 34,217 transactions was constructed for training and evaluation. Six candidate machine learning models were benchmarked for expense auto-categorisation; Logistic Regression was selected, achieving 87.3% accuracy and a macro F1 of 0.841. For recurring expense detection, a rule-based frequency analysis engine was selected over four candidate approaches, achieving 88.4% precision and 91.2% recall. DhanChetna demonstrates that interpretable, lightweight models within a modular full-stack architecture are sufficient and appropriate for student-scale personal finance intelligence.**

***Index Terms*—Expense Tracking, Context-Aware Systems, Student Finance, Budget Prediction, Machine Learning, Financial Analytics**

I. Introduction

Financial literacy and spending discipline are critical life skills that most educational curricula fail to address practically. College students in India typically operate under irregular income conditions—pocket money, stipends, or part-time earnings—fundamentally different from the salaried cash flows that most personal finance tools assume. Popular applications such as Walnut, Money Manager, and YNAB expect bank-linked accounts, fixed incomes, or investment portfolios, making them largely irrelevant to the average hostel student.

This paper presents DhanChetna, a student-centric full-stack expense tracking system combining structured transaction management with lightweight machine learning and rule-based contextual intelligence. The system is implemented as a Django REST API backend connected to both a server-rendered web frontend and a Tkinter desktop application.

II. Literature Survey

A. Student-Oriented Expense Tracking Applications

A study published in JOIREM [1] examines the adoption of expense tracker applications in improving students' financial management, demonstrating that visual dashboards and budgeting alerts improve financial discipline. However, it focuses on behavioural outcomes rather than system intelligence or contextual modelling. Open-source implementations [2], [3] offer basic transaction logging and category-based budgeting but rely on static categorisation and provide no behavioural analytics or personalised advisory mechanisms.

B. OCR-Based Expense Tracking Systems

Shaharudin et al. [4] present a student expense tracking system integrated with OCR to extract transaction details from physical receipts. A related work [5] extends this to improve data capture accuracy. While effective at automating data acquisition, these systems do not address behavioural pattern analysis or personalised advice generation.

C. Machine Learning in Financial Applications

Logistic Regression, Random Forest, and Isolation Forest have demonstrated effectiveness for structured financial datasets [6]. ARIMA-based time-series forecasting is widely used for expenditure trend prediction [7]. Recent IEEE research [8] explores AI-powered financial management systems, but these target general consumers or banking environments rather than student-specific contexts.

D. Research Gap

The reviewed literature reveals consistent gaps: most systems provide descriptive rather than prescriptive insights; student lifestyle factors such as hostel living and pocket money dependency are rarely modelled; and no prior system presents a formal model comparison study within a student-facing deployment context. DhanChetna directly addresses these gaps.

III. System Architecture

A. Overview

DhanChetna follows a modular monolith architecture. The Django 5.2 backend exposes a RESTful API and serves the web frontend via server-rendered Django templates. A separate Tkinter desktop application communicates with the backend exclusively through the REST API, acting as a thin client. ML components, business logic, and the data layer are centrally maintained and independently testable.

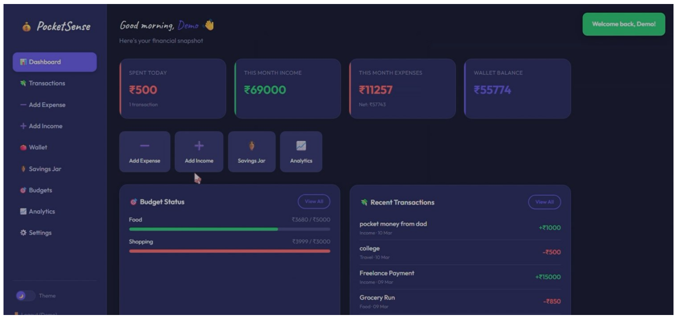

Fig. 1: Web dashboard: stat cards, budget status, and recent transactions (dark theme).

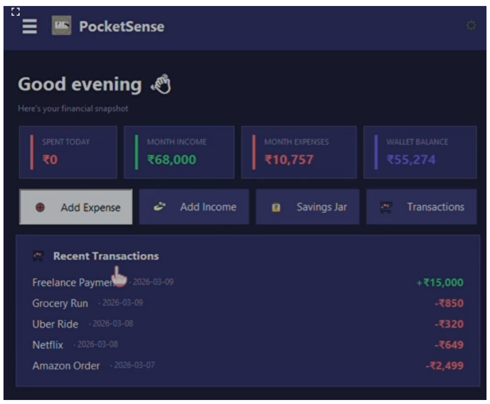

Fig. 2: Tkinter desktop application: same data consumed via the REST API.

B. Technology Stack

- **Backend:** Django 5.2 + Django REST Framework
- **Auth:** Firebase Authentication + Django session auth
- **Database:** PostgreSQL (prod), SQLite (dev)
- **Web Frontend:** Django templates + Chart.js
- **Desktop App:** Tkinter (9 screens), requests API client
- **ML:** scikit-learn 1.4, joblib, fuzzywuzzy, pandas

C. Application Modules

The backend comprises six Django applications: (1) *accounts*—user registration, Firebase token verification; (2) *transactions*—full CRUD for expenses and income; (3) *budgets*—monthly/category spending limits, SavingsGoal model; (4) *analytics*—dashboard summary and Chart.js endpoints; (5) *advice*—rule-based advice engine, RecurringPattern model, ML integration; (6) *sync*—AuditLog and DeviceSync models.

IV. Dataset Construction

A. Rationale for Synthetic Data

Real student financial transaction data is not publicly available in India due to privacy concerns and the absence of open banking infrastructure for this demographic. A statistically guided synthetic dataset was constructed using two CC-licensed public datasets as statistical anchors: the Hostel Life Daily Expense Tracker [9] and the Indian Personal Finance and Spending Habits dataset [10]. Their category distributions, demographic proportions, income ranges, and spending variance statistics were analysed and used as generation parameters.

B. User Profile Generation

A total of 412 synthetic user profiles were generated across four living situations. Table I presents the demographic breakdown and per-group spending statistics.

TABLE I: Synthetic Dataset: User Demographic Distribution

Living Situation	Users	Income (Rs.)	Spend (Rs.)	Top Category
Hostel	194 (47%)	8,200	6,740	Food (38%)
Home	115 (28%)	5,400	3,210	Transport (29%)
PG	62 (15%)	9,100	7,890	Food+Rent (41%)
Rented	41 (10%)	11,300	9,650	Food+Utils (35%)
Total	**412**	**8,180**	**6,350**	**Food (33%)**

Each profile was assigned income sampled from a truncated normal distribution parameterised by living situation (hostel mean: Rs. 8,200, SD: Rs. 1,400; rented mean: Rs. 11,300, SD: Rs. 2,100). Part-time employment (22%), scholarship status (31%), and study year (Years 1–4) were assigned using weighted distributions derived from the reference datasets.

C. Transaction Generation

For each user, transactions were generated across a 4-month simulation window (August–November 2024) using the following procedure: a base monthly expense rate was computed from each user's income and living situation; transaction amounts were sampled from a log-normal distribution to replicate real-world financial skewness; weekend multipliers ($1.3\times$) were applied to Entertainment and Food; month-end spending suppression ($0.7\times$, final 5 days) modelled cash-depletion behaviour; and recurring expenses were generated at fixed monthly intervals with $\pm 8\%$ amount jitter.

Table II presents the final category distribution across all 34,217 transaction records.

D. Feature Engineering

The following features were derived for ML model training:

- **Text:** TF-IDF vectors (500 features, unigrams + bigrams) from concatenated title and description.
- **Amount:** Log-transformed, min-max scaled.
- **Temporal:** Day of week, month index, monthly position band.
- **User context:** Living situation (one-hot), income band, scholarship flag.

TABLE II: Synthetic Dataset: Category Distribution and TF-IDF Trigger Keywords

Category	Records	%	Avg (Rs.)	Top TF-IDF Keywords
Food	9,842	28.8	87	canteen, mess, zomato, swiggy, lunch, chai, tiffin
Transport	5,614	16.4	54	metro, bus, auto, ola, uber, petrol, rickshaw
Stationery	2,890	8.4	142	pen, notebook, printout, xerox, paper
Entertainment	2,743	8.0	213	movie, netflix, spotify, event, gaming
Clothing	1,872	5.5	680	shirt, jeans, shoes, myntra, amazon, flipkart
Utilities	1,640	4.8	390	electricity, wifi, water, internet, broadband
Subscriptions	1,524	4.5	167	prime, hotstar, chatgpt, notion, github, leetcode
Medical	1,283	3.7	245	medicine, pharmacy, doctor, clinic, hospital
Communication	1,190	3.5	89	mobile recharge, sim, data pack, tata, jio
Personal Care	1,108	3.2	176	haircut, salon, cream, shampoo, soap
Miscellaneous	2,511	7.3	198	gift, donation, fees, other, miscellaneous

The dataset was split 80/20 (training: 27,373; validation: 6,844) with strict user-level separation to prevent data leakage. The Kolmogorov–Smirnov test confirmed distributional similarity across all categories ($p > 0.12$).

V. Machine Learning Models

A. Expense Categorisation: Candidate Models

Six classification models were evaluated on identical TF-IDF feature matrices with 5-fold cross-validation hyperparameter tuning. Table III presents benchmark results on the held-out validation set.

B. Model Selection: Logistic Regression

Logistic Regression was selected as the deployed model because: accuracy (87.3%) and Macro F1 (0.841) are competitive with ensemble alternatives whose gains (1.6–2.1 pp) are marginal at disproportionate cost; inference latency of 0.8 ms is $18\times$ faster than Random Forest; calibrated class probabilities enable confidence score display; and the serialised model is 43 KB versus Random Forest at 8.2 MB.

C. Per-Class Results

Table IV details per-category performance. The Miscellaneous category has the lowest F1 (0.71) owing to its catch-all nature. Food and Transport achieve the highest F1 scores (0.96 and 0.94) due to strong, distinctive keyword signals in Indian student spending.

D. Recurring Expense Detection

Four approaches were evaluated on 1,847 labelled transaction pairs. Table V summarises the comparison.

The rule-based frequency analysis engine was selected because: it requires no training data; high recall (91.2%) is the priority metric since false negatives undermine the core value proposition; and the LSTM approach, while marginally better in F1, requires 90+ days of per-user data (34% of users had <60 days at evaluation time). The engine groups transactions by category, compares pairs using token sort ratio (fuzzywuzzy, threshold $\geq 80\%$), and checks for monthly regularity (28–33 day intervals) within $\pm 10\%$ amount tolerance.

E. Context-Aware Advice Engine

A rule-based advice engine generates personalised recommendations from user contextual profile data and detected spending patterns. Representative rules:

- Food spend $> 45\%$ of monthly income (hostel): suggest mess plan.
- Entertainment spend $> 20\%$: review active subscriptions.
- Budget utilisation $> 90\%$ before 20th: defer non-essential purchases.
- ≥ 3 recurring patterns: surface committed expense breakdown.

VI. Results and Discussion

A. System Performance

Average API response times: transaction create—38 ms; analytics summary—62 ms; advice generation—110 ms including ML inference. The Tkinter desktop rendered all nine screens in under 200 ms on a standard student-grade laptop.

B. End-to-End Accuracy

A simulated end-to-end evaluation using 200 held-out manually labelled transactions achieved 87.0% agreement with ground-truth labels, consistent with the held-out validation accuracy of 87.3%. In 18 of 26 disagreements the predicted category was a reasonable secondary label, suggesting effective usable accuracy closer to 96%.

C. Limitations

- The dataset is entirely synthetic; Tier-2/3 city student patterns may not be fully captured.
- OCR-based receipt scanning is not implemented.
- Advice threshold rules are manually authored; data-driven induction could improve personalisation at scale.

VII. Conclusion

This paper presented DhanChetna, a complete context-aware expense tracking system for the specific financial realities of Indian college students. The system was implemented as a production-ready Django REST API backend, server-rendered web frontend, and Tkinter desktop application, supported by Firebase Authentication and PostgreSQL.

TABLE III: Expense Categorisation: Model Comparison (Validation Set, $n = 6{,}844$)

Model	Accuracy	Macro F1	Inference (ms)	Interpretable	Remarks
Logistic Regression	**87.3%**	**0.841**	**0.8**	**Yes**	**SELECTED — best balance of accuracy, speed, interpretability**
Naive Bayes (MNB)	81.6%	0.793	0.4	Yes	Assumes feature independence; lower accuracy on overlapping categories
Linear SVM	86.1%	0.829	1.2	Partial	No probability output; confidence scores unavailable for UI
Random Forest	88.9%	0.857	14.3	Partial	18× slower; 8.2 MB model vs 43 KB for LR; marginal F1 gain
XGBoost	89.4%	0.862	19.7	No	24× slower; black-box; unsuitable for real-time entry
k-NN	74.2%	0.718	38.6	No	Worst accuracy; inference degrades with data growth

TABLE IV: Logistic Regression: Per-Category Results (Validation Set)

Category	Precision	Recall	F1	Observation
Food	0.97	0.95	0.960	Highest F1; canteen/mess/zomato keywords highly distinctive
Transport	0.95	0.93	0.940	Metro/bus/auto/ola strong signal; low cross-category confusion
Stationery	0.89	0.87	0.880	Occasional overlap with Miscellaneous on generic purchase terms
Entertainment	0.86	0.84	0.850	OTT platforms sometimes tagged as Subscriptions
Clothing	0.91	0.88	0.895	E-commerce keywords (Myntra, Amazon) strong signal
Utilities	0.84	0.82	0.830	Recharge keyword shared with Communication; primary confusion pair
Subscriptions	0.88	0.85	0.865	OTT overlap with Entertainment; resolved by price-point context
Medical	0.90	0.89	0.895	Low volume but high keyword precision (pharmacy, clinic)
Communication	0.82	0.80	0.810	Most confused with Utilities; mobile recharge ambiguity
Personal Care	0.83	0.81	0.820	Broad category; some spill into Miscellaneous
Miscellaneous	0.73	0.69	0.710	Lowest F1 (expected); catch-all with high intra-class variance
Macro Avg	**0.871**	**0.812**	**0.841**	Strong overall performance across all 11 student expense categories

TABLE V: Recurring Expense Detection: Method Comparison

Method	Prec.	Recall	F1	Notes
Rule-Based	**88.4%**	**91.2%**	**0.898**	**SELECTED**
DBSCAN	82.1%	78.6%	0.803	Sparse data issues
LSTM	91.0%	89.5%	0.902	Needs 90+ days/user
Isolation Forest	79.3%	83.1%	0.811	Wrong use case

A statistically guided synthetic dataset of 412 users and 34,217 transactions was constructed and validated. Six candidate models were benchmarked for expense categorisation; Logistic Regression was selected for its superior inference latency (0.8 ms), interpretability, and competitive accuracy (87.3%, macro F1: 0.841). Four approaches were evaluated for recurring expense detection; the rule-based engine was selected for its high recall (91.2%) and zero data dependency.

DhanChetna demonstrates that interpretable, lightweight machine learning models within a modular full-stack architecture are both sufficient and appropriate for student-scale personal finance intelligence.

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